

TikTok Project Final Report

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Introduction

This report combines the main results from the previous assignments. We first give a short overview of the data, then summarize the main descriptive analyses, and finally present the regression analysis.

Data Overview

The project uses TikTok-related data from both a SQLite database and several CSV files. Across the assignments, we worked with information on videos, impressions, sessions, users, and watch events.

Video overview

```
  n_videos n_creators avg_length avg_watch_rate
1    50000         500   30.12118    0.6940614
```

Short vs long videos

```
# A tibble: 2 x 4
  length_group videos_n watch_rate_avg watch_share_avg
  <chr>          <int>         <dbl>         <dbl>
1 long           26030         0.695         0.378
2 short          23970         0.693         0.376
```

Short and long videos had very similar average watch rate and watch share. So based on this descriptive comparison, video length alone does not seem to make a clear difference in audience retention.

Descriptive Analyses

Impressions

The impressions dataset originally contained 95,737 rows. We found 786 fully duplicated rows and removed them, leaving 94,951 rows. The `mission_ids` variable also had 13,926 missing values.

After removing the exact duplicates, 161 repeated `impression_id` values were still present. These were kept because the other variables differed, meaning they likely represented separate impressions rather than data errors.

Most impressions came from the `preferred_new` feed (51.6%), followed by `explore` (31.4%) and `known` (17.0%).

Session analysis

We also explored how users behave within sessions. This part of the analysis looked at session duration, the number of videos viewed, watch time, and how sessions changed over time.

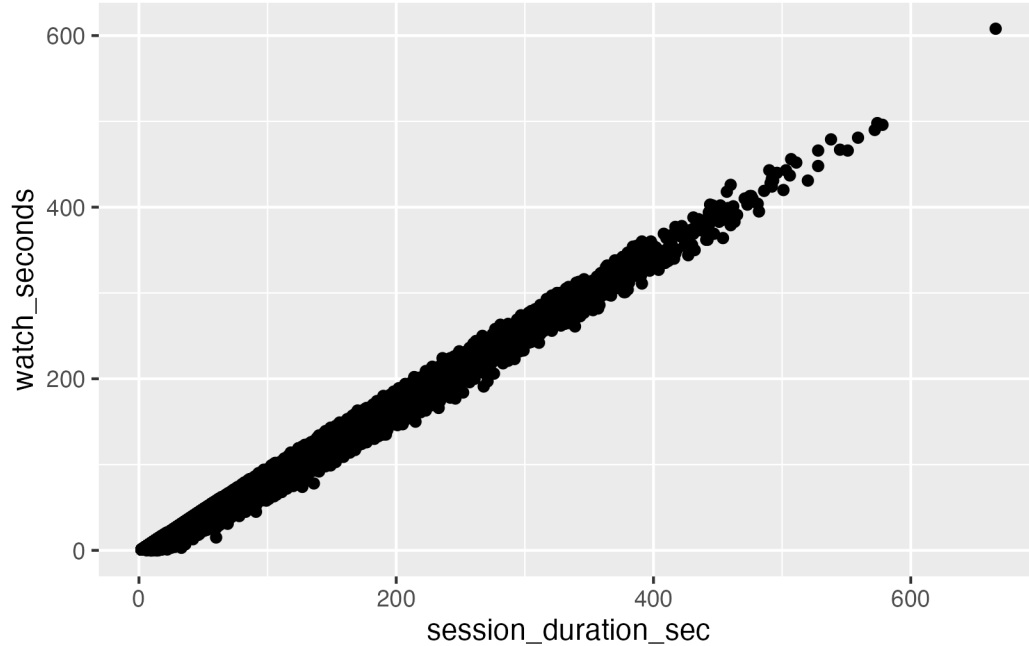


Figure 1: Relationship between session duration and watch time

The session analysis was mainly exploratory. It gave a general picture of how long sessions last, how many videos users watch in one session, and how watch time relates to session behaviour.

User analysis

The users dataset was cleaned by removing rows with missing values and duplicate observations.



Figure 2: Beauty/Fashion preference and mean videos watched

We compared users with different preferences, especially Beauty/Fashion and Gaming. We also looked at differences in satiation decay, need for interaction, and the relationship between Beauty/Fashion preference and the average number of videos watched.

Watch events

The watch events analysis focused on user actions, the relationship between video length and watch time, and the number of watch events over time.

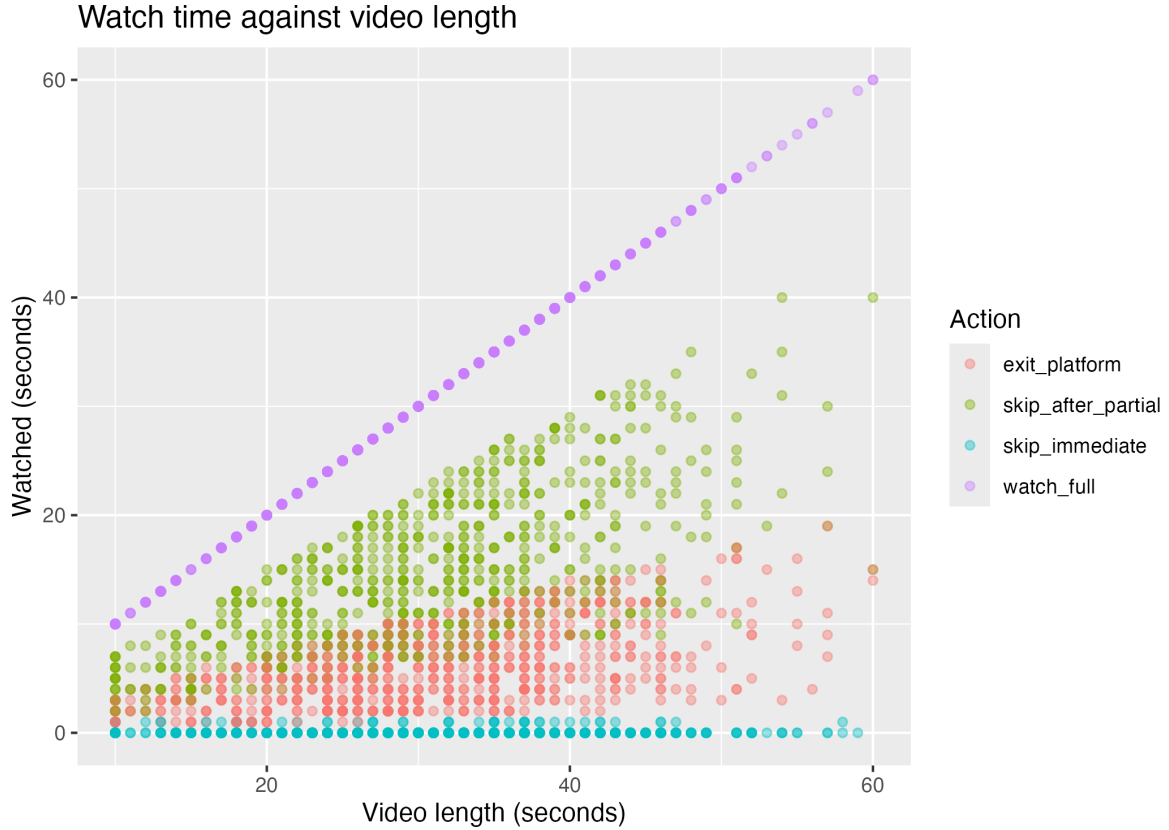


Figure 3: Watch time against video length

This part of the analysis gives a descriptive view of how users react to videos and how watch time changes with video length.

Regression Analysis

The final part of the project studies whether having seen a creator before is related to watch time.

The regression uses data from `watch_logs` and `video_view`. Two main variables are created: `seen_before`, which indicates whether a user has already seen the same creator, and `long_video`, which indicates whether a video is longer than the median video length.

We estimated four models: - Model 1: watch time explained by prior creator exposure - Model 2: adds video length - Model 3: includes the interaction between prior exposure and video length - Model 4: repeats the interaction model using log-transformed watch time

Model output

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.4964535	0.01547606	742.85408	0.000000e+00
seen_before	-0.3045701	0.01894801	-16.07399	3.915721e-58

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.7518997	0.01722180	508.18717	0.000000e+00
seen_before	-0.3146063	0.01845187	-17.05011	3.52644e-65
long_video	5.7315043	0.01740477	329.30647	0.000000e+00

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.7162814	0.02087641	417.51813	0.000000e+00
seen_before	-0.2611540	0.02557424	-10.21160	1.761795e-24
long_video	5.8058867	0.03016854	192.44836	0.000000e+00
seen_before:long_video	-0.1114899	0.03693491	-3.01855	2.539909e-03

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.64028636	0.002289679	716.3828020	0.000000e+00
seen_before	-0.03095010	0.002804926	-11.0341981	2.618680e-28
long_video	0.33916142	0.003308819	102.5022713	0.000000e+00
seen_before:long_video	0.00154471	0.004050938	0.3813216	7.029646e-01

Regression figures

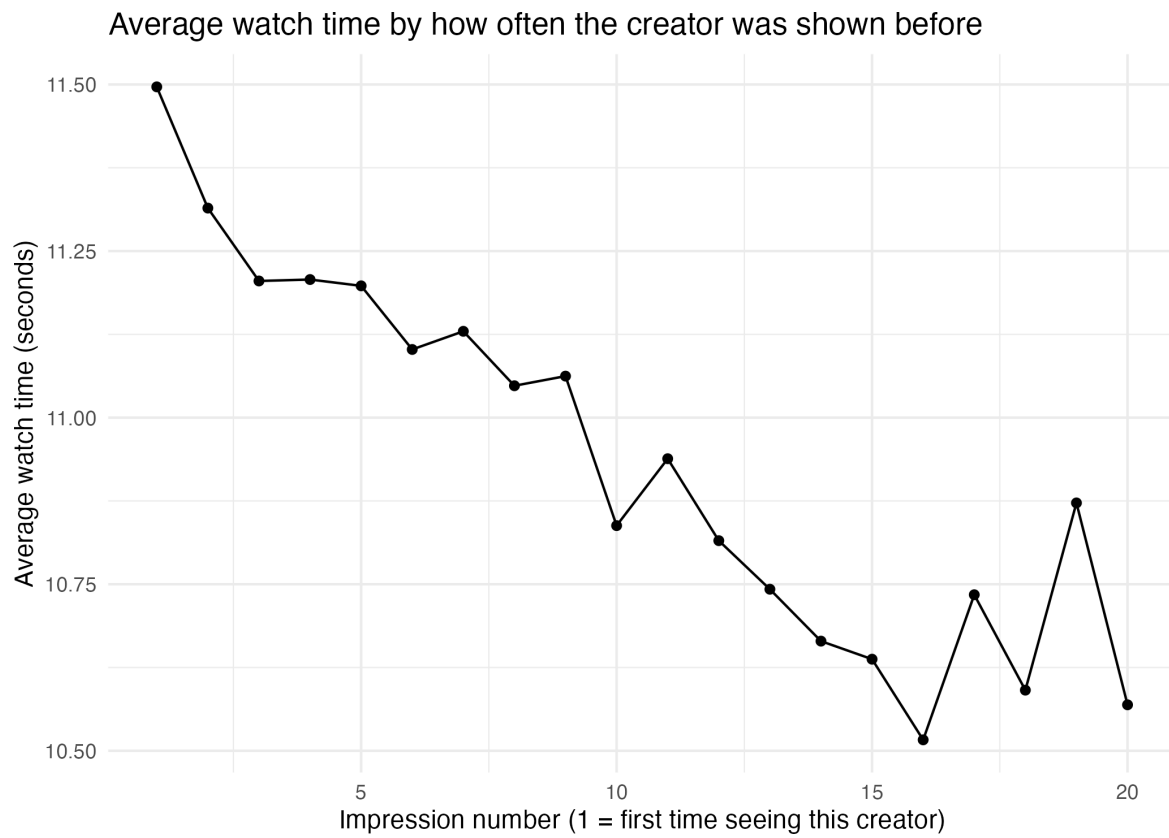


Figure 4: Average watch time by prior exposure to a creator

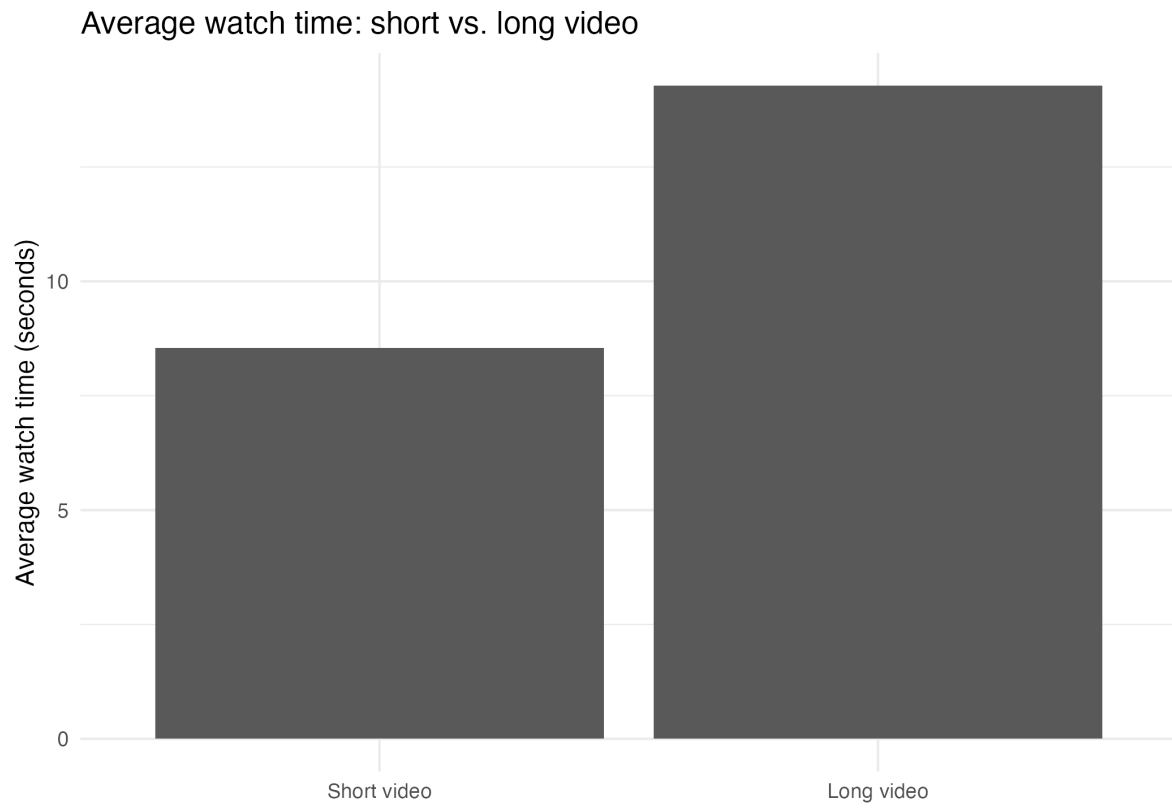


Figure 5: Average watch time for short and long videos

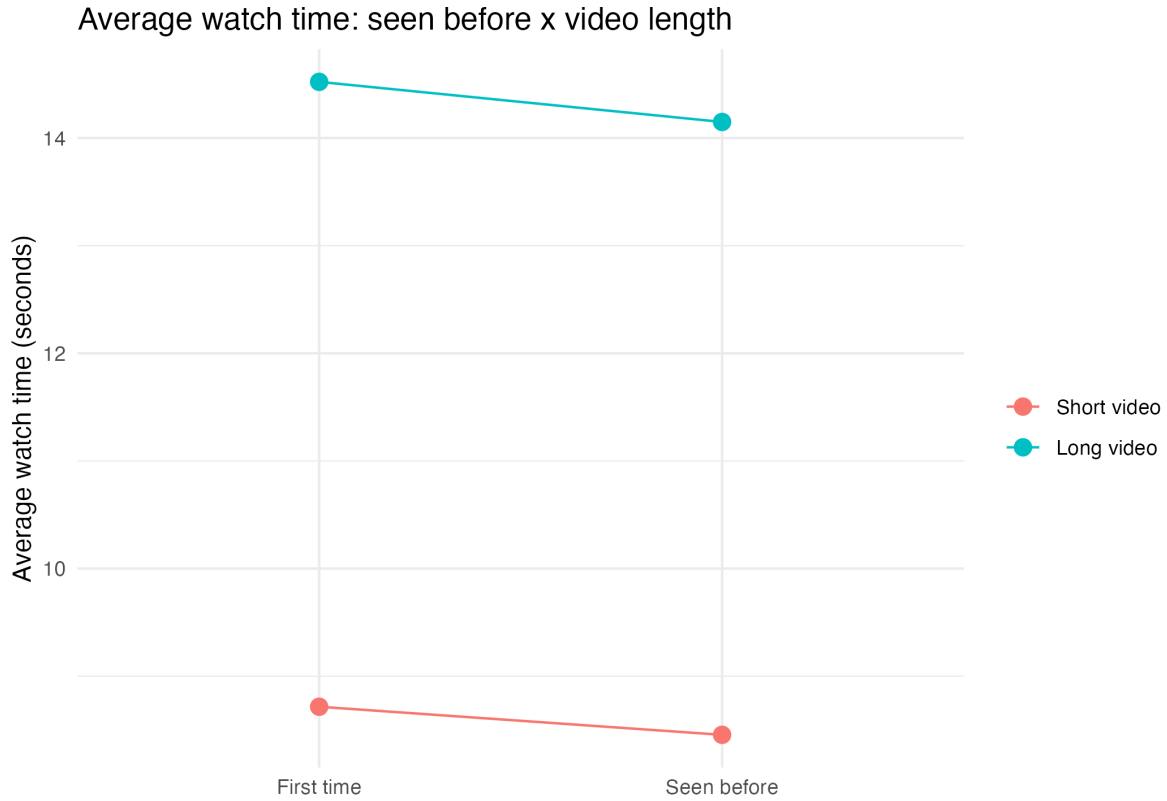


Figure 6: Interaction between prior exposure and video length

Regression Results

In Model 1, **seen_before** has a negative coefficient of -0.305. This means that users watched about 0.3 seconds less, on average, when they had seen the same creator before. The effect is statistically significant, although it is very small.

After adding video length in Model 2, the coefficient for **seen_before** remains negative (-0.315). Long videos are associated with about 5.73 seconds more watch time than short videos.

Model 3 includes an interaction between prior exposure and video length. The interaction coefficient is negative (-0.111) and statistically significant. This suggests that the negative relationship between prior creator exposure and watch time is slightly stronger for long videos.

However, in Model 4, where watch time is log-transformed, the interaction term is no longer significant ($p = 0.703$). This means that the interaction effect is not very robust across model specifications.

Conclusion

The descriptive analyses showed several patterns in TikTok viewing behaviour across videos, impressions, sessions, users, and watch events.

The regression analysis showed that previous exposure to the same creator was associated with slightly lower watch time, while longer videos were associated with higher watch time. The interaction between these two variables was statistically significant in the model using raw watch time, but not in the log-transformed model.

The size of the prior-exposure effect was small, and the models explained only a limited part of the variation in watch time. The results therefore show associations in this dataset rather than strong evidence of a causal relationship.

Limitations

This project has a few limitations. Some datasets needed cleaning, and some variables contained missing values or duplicate records. Also, not all potentially important factors are observed in the data.

Because of this, the results should be understood as a structured analysis of the available data rather than a complete explanation of user behaviour.